NUCLEAR ENGINEERING AND DESIGN 26 (1974) 444–460. © NORTH-HOLLAND PUBLISHING COMPANY

LITHIUM HYDRIDE: A SPACE AGE SHIELDING MATERIAL*

Frank H. WELCH
Atomics International Division, Rockwell International Corporation, Canoga Park, California 91304, USA

Received 7 May 1973

Lithium hydride (LiH) is a useful, lightweight, neutron-shielding material for mobile nuclear reactors, such as the one considered for the Space Reactor Program. The fabrication of LiH bodies is traced from early pressing and casting operations through over a decade of casting development work to the current casting capability existing at the Nuclear Field Laboratory of Atomics International. The capability of the newest LiH shield casting facility, and what is bielieved to be the world's largest LiH neutron shield are discussed in detail. Basic characteristics of a number of LiH neutron shields are also tabulated.

1. Introduction

Men and materials performing in the environment of an operating nuclear reactor require shielding from the escaping neutron particles and gamma rays. For efficient shielding from gamma rays, dense, high atomic number elements, such as iron, lead, or tungsten are required; whereas light, low atomic number elements, such as hydrogen, lithium, or beryllium are required for efficient neutron shielding.

This paper considers only the use of lithium hydride (LiH) as a highly efficient neutron-shielding material. It contains, combined into a single, stable compound, two of the elements most effective in attenuating and absorbing neutrons.

2. Lithium hydride

Ground-based reactors utilize concrete, water, or other hydrogen-containing materials to provide neutron shielding in an application where the total weight of shielding is relatively unimportant. However, the mobile nuclear reactor, such as that envisaged in the now-defunct Nuclear Aircraft Program and Space Reactor Program*, is extremely weight-sensitive and requires the use of a high hydrogen content, low density material.

Many liquid and solid, hydrogen-containing compounds were evaluated and studied in the laboratory before LiH was selected as the most effective shiel ding material. The unique properties of LiH, as listed by Welch [1, 2], which led to this choice were:

(1) high hydrogen content (12.68 wt % H in LiH) and high N_H, the number of H atoms/cc (5.85×10^{22}) which provide a highly efficient means of shielding men and equipment;

(2) high melting temperature (686°C) which permits the material to be used in the high thermal environs of the reactor without melting;

(3) low dissociation pressure (~ 20 torr at the melting point) which permits the material to be melted and frozen without degradation under a slight pressure of hydrogen; and

(4) low density (0.775 g/ml at room temperature) which makes it attractive for use in a space system.

The material properties which are unattractive and may complicate a shield design are its low thermal conductivity and poor mechanical properties. Generally, these characteristics do not detract from its use.

* This work was conducted at Atomics International under the auspises of the USAEC and NASA. It comprises a section from chapter 57, 'Hydrides as Shielding Materials', of the proposed third edition of the *USAEC Reactor Materials Handbook*, revised and updated. See also the paper by R. Van Houten, in this issue, and that by C.D. Montgomery in *Nuclear Engineering and Design* 25, 309.

* USAEC and NASA program cancellation occurred in early 1973.

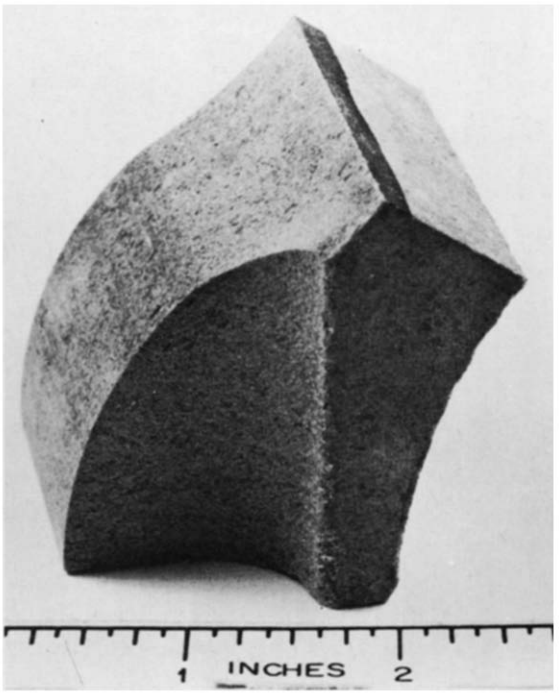


Fig. 1. Cold pressed and machined LiH shape (GE-ANPD program).

Raw material quality and quantity are quite favorable; ton quantities are available from several commercial sources at a purity of 97.5%. The cost of commercial LiH is high ($ 12–13/lb), but its low density and high hydrogen content offset the high cost.

2.1. Fabrication

Lithium hydride shapes can be fabricated by cold pressing, hot pressing, and melting and casting. Regardless of the fabrication method, LiH must be protected at room temperature from water and water vapor, and numerous water-containing and chemically active materials. At elevated temperatures, LiH also must be protected from oxygen, nitrogen, carbon dioxide, and many other elements and compounds. In addition, LiH must be protected at elevated temperatures by a slight hydrogen overpressure, or by a sealed container, to prevent the loss of hydrogen by decomposition.

Cold pressing and machining of irregular blocks are reported in refs. [3] and [4] for the fabrication of LiH test specimens. Cold pressing and machining

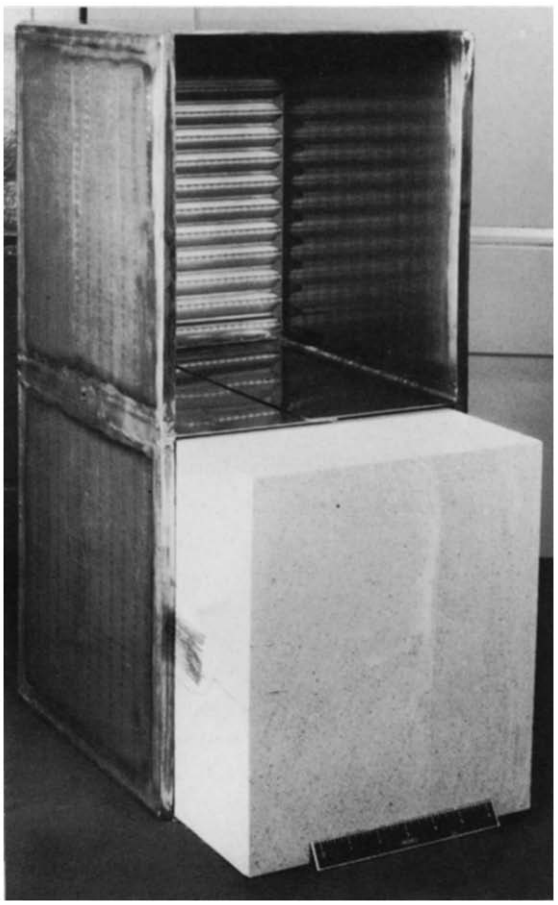


Fig. 2. Large cube of cold pressed and machined LiH (GE-ANPD program).

of LiH shapes were utilized also in the early fabrication of shields for both the Aircraft Nuclear Propulsion (ANP) Project (figs. 1 and 2) and the Space Nuclear Auxiliary Power (SNAP) Program, as documented in refs. [5] and [6]. Both programs abandoned this technique when prohibitive difficulties arose during studies of retaining the pressed blocks in the shield vessel in a manner which would accommodate the differential thermal expansion between the steel and the LiH, yet prevent shock and vibration loads from damaging either the vessel or the LiH.

Small specimens of LiH have been hot pressed at 380°C and 10 000 psi to about 98% of the theoretical density [3], and at 500–600°C and 2000 psi to near the theoretical density of 0.775 g/ml [7]. No

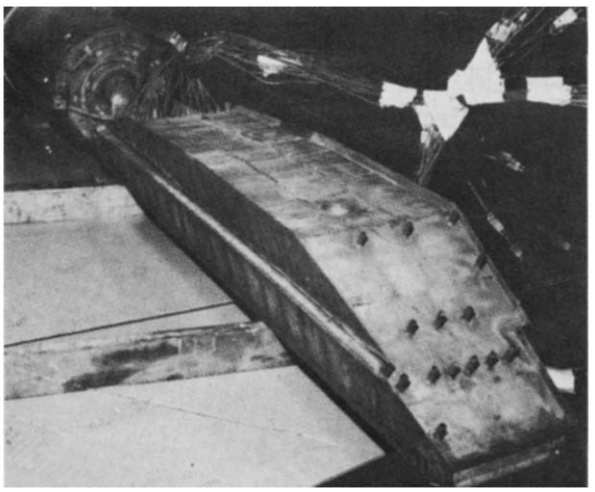


Fig. 3. Six foot long LiH casting produced at GE-ANPD.

information exists on the hot pressing of large LiH shapes.

Several potential methods are discussed by Woods [4] for the fabrication of LiH shapes for the Sunflower Project. The direct hydriding of lithium metal in a metallic container is presented as a promising technique.

2.1.1. Casting and its history

Several early castings of LiH are reported in refs. [8] and [9], but the work was not directed towards producing neutron shielding. Aitken [10] described the casting process as a method of fabricating LiH neutron shielding at the General Electric-ANP, Department, Cincinnati, Ohio. Several laboratory-size castings were made to establish the feasibility of the process, including a casting into a welded, stainless steel honeycomb matrix. Zone cooling of the LiH by withdrawal of the container from the furnace was demonstrated to be a useful technique.

A development program was initiated at the Union Carbide Nuclear Company (UCNC) Y-12 Plant, Oak Ridge, Tenessee by GE-ANP to investigate further the LiH casting process. The program was quite successful in demonstrating a number of different casting techniques, and in fabricating cast specimens in a variety of complex shapes [5, 11]. Among the techniques investigated were:

(1) Transfer casting – molten LiH was transferred from a heated vessel to the casting container using gravity flow or inert gas pressure to transfer the liquid.

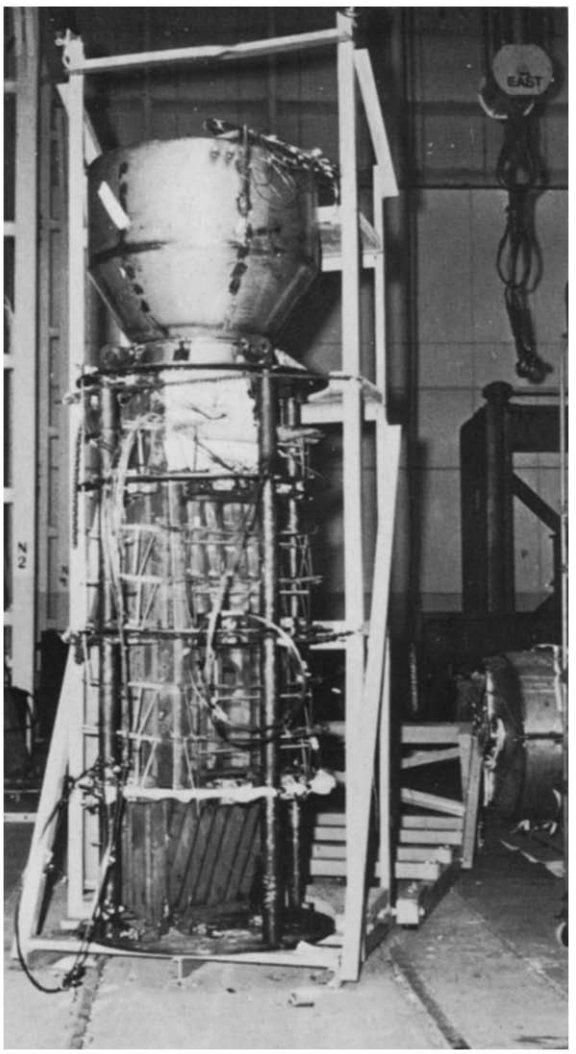


Fig. 4. Nine foot long LiH casting produced at GE-ANPD.

(2) Zone melting – a heated furnace was moved slowly up a LiH-filled cylinder causing the molten zone to move up the cylinder, similar to a zone refining operation.

(3) Zone cooling – a container filled with molten LiH was directionally cooled to promote freezing from the bottom to the top, either by the stationary method, whereby the furnace heat is controlled to produce a temperature gradient, or by the withdrawal method whereby the casting container is slowly lowered out of the heated furnace.

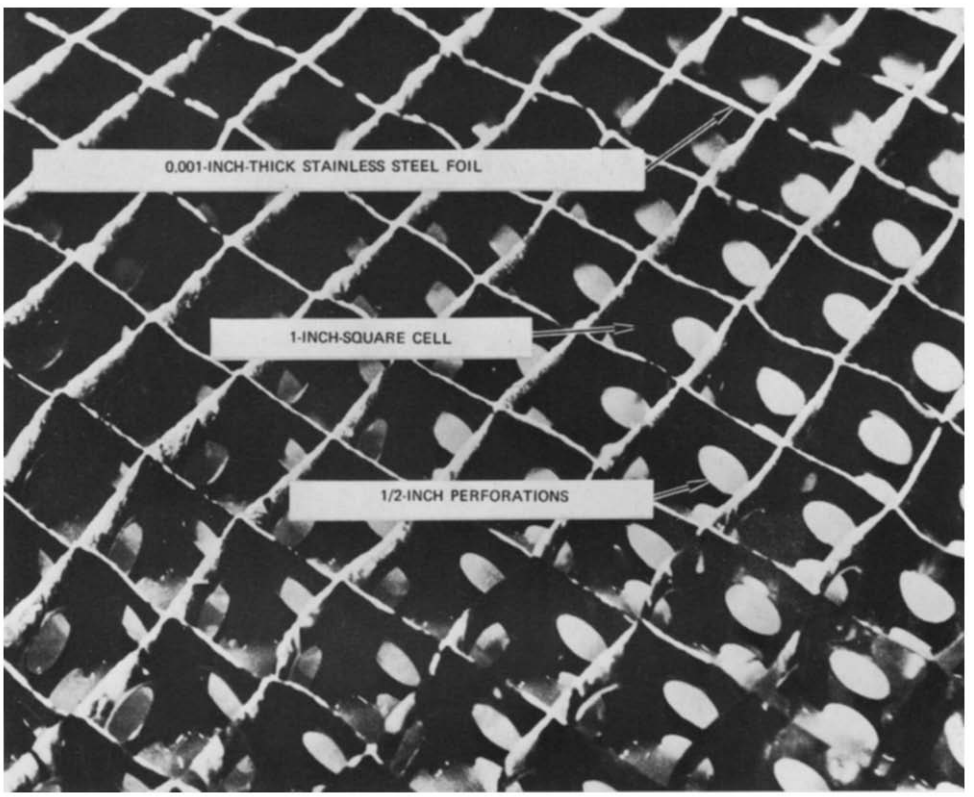


Fig. 5. Close-up of honeycomb matrix.

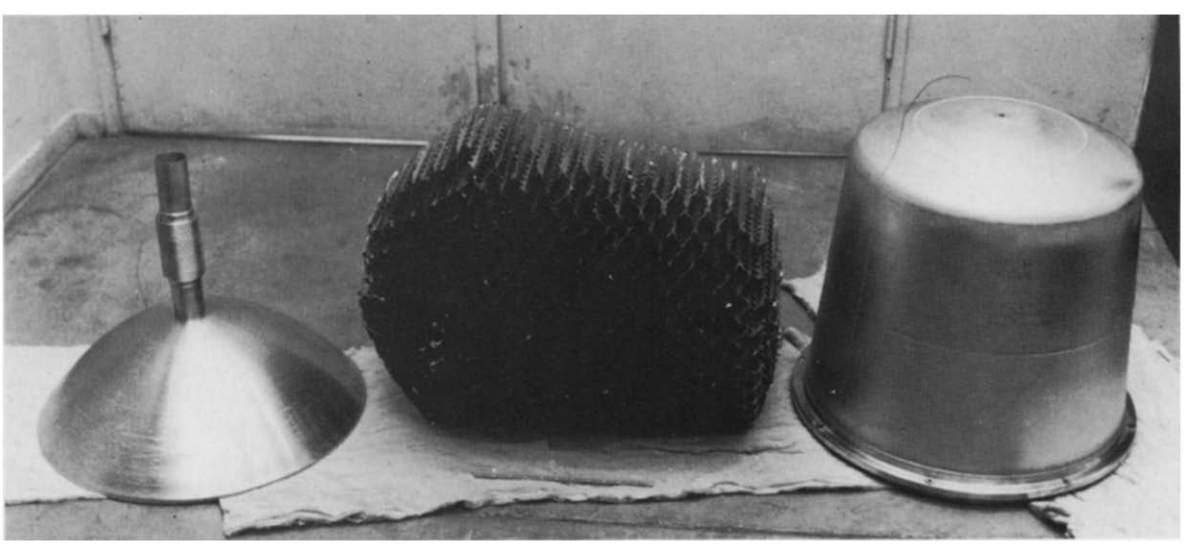


Fig. 6. Shield vessel, shaped honeycomb block, and vessel bottom.

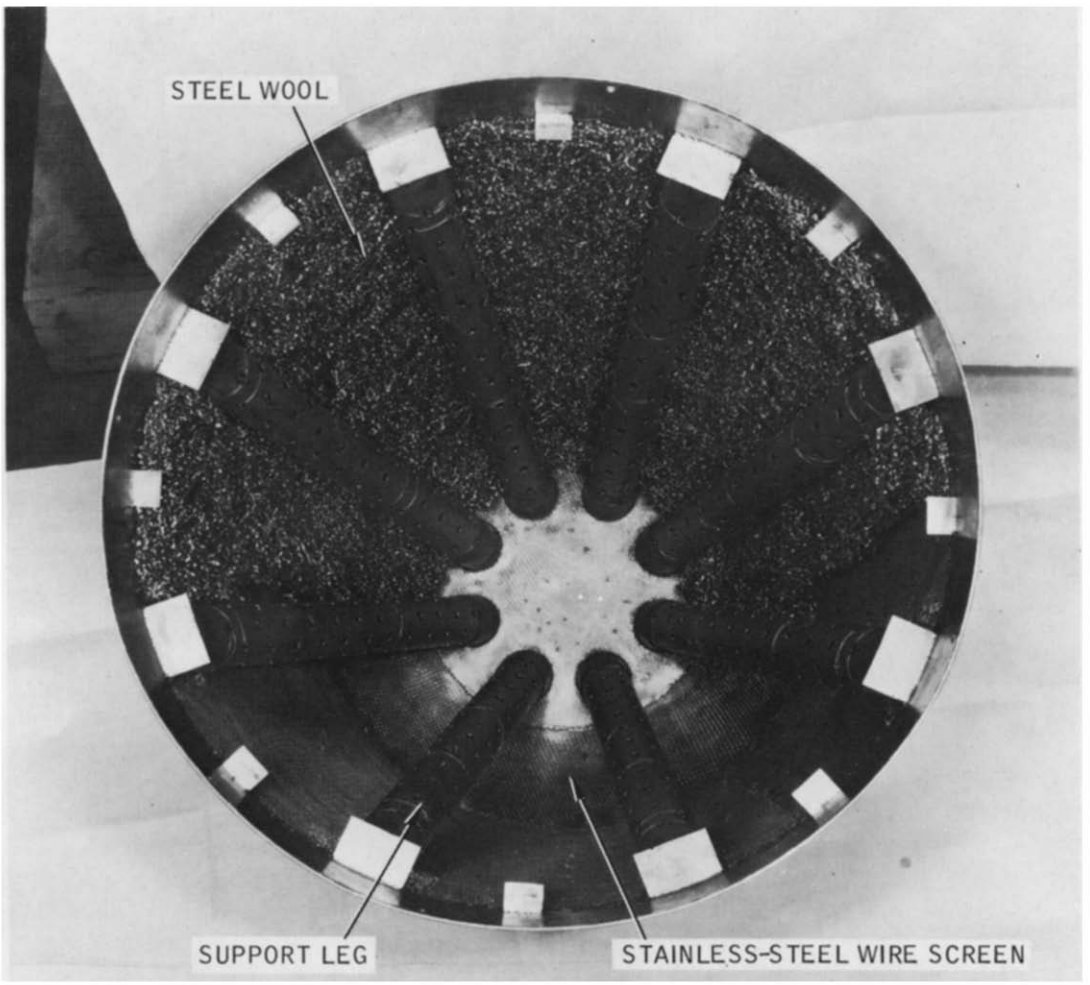


Fig. 7. Interior of experimental shield no. 1 showing steel wool and support spider.

Baxter [5] reports that the success of this development program led to the initiation of a more comprehensive program at GE-ANP which produced sound, cast specimens repetitively, although equipment limitations restricted the length of the specimens to about 3ft. Baxter [12] further reports that programs were begun simultaneously at GE-ANP and UCNC to produce larger LiH castings. The latter program produced a number of castings 12 in. in dia. by 54 in. long using the stationary method, but the program was terminated after the successful casting of 6 ft long specimens at the GE-ANP facility. Fig. 3 shows a typical 6ft long specimen, fabricated by a modified zone cooling-withdrawal process, whereby the furnace was raised stepwise to permit the cooling to occur from the bottom to the top.

The GE-ANP program was continued with the construction of a larger facility possessing a casting retort of 5 ft dia. and $19\frac{1}{2}$ ft high [13]. Three full-sized shield sectors (fig. 4), each containing about 900 lb of LiH were cast. Two castings were successful, but the third specimens developed leaks which resulted in gross loss of LiH. In general, the LiH casting program was believed to be quite successful, as noted by Poor [14].

The development of the successful LiH casting techniques at GE-ANP introduced the possibility of a new fabrication method for SNAP shields which

Table 1.
Characteristics of cast LiH shields and test specimens.

Shield/ specimen	Program	Description	Total volume (ft^3)	Void volume (ft^3)	% dense	LiH density (lb/ft^3)
SNAP 10A test casting [15]	USAEC *	cylindrical vessel containing 200 lb LiH and two unmachined honeycomb blocks (fig. 5)	4.32	0.24	94.4	45.7
SNAP 10A cast shield [15]	USAEC	conical vessel containing 200 lb LiH and a machined honeycomb (fig. 6)	4.32	0.29	93.3	45.1
SNAP 8 thermal shield [15]	USAEC	conical SNAP 10A vessel with hemispherical head containing 240 lb LiH and a machined honeycomb shape, instrumented with 15 internal thermocouples	5.15	0.29	94.3	45.6
SNAP 8 two-region test casting [15]	USAEC	cylindrical vessel containing 450 lb LiH	9.41	0.67	92.9	44.9
SNAP 8 two-region DRM-1 shield [15]	USAEC	conical vessel containing 720 lb LiH and a machined block of honeycomb in each region	14.93	0.92	93.8	45.4
SNAP 10A DRM-1C developmental shield [15]	USAEC	conical SNAP 10A vessel containing 200 lb LiH, a machined honeycomb block, and a lightweight internal rib structure	4.32	0.22	95.0	46.0
SNAP 8 experimental shield no. 1 [15]	USAEC	conical SNAP 10A vessel containing 200 lb LiH, a steel wool filler, and a light-weight internal tubular support for the reactor loads (fig. 7)	4.32	0.24	94.2	45.6
SNAP 2 develop-mental shield [15]	USAEC	conical vessel containing 300 lb LiH and heavy internal ribs on the cone to support reactor loads (fig. 8)	6.27	0.40	93.6	45.3
SNAP 2 DRM-1 shield [15]	USAEC	same as above	6.27	0.40	93.6	45.3
Experimental shield no. 2 [16]	USAEC	conical SNAP 10A vessel containing 200 lb LiH, a 200 lb layer of granulated tungsten alloy infiltrated with LiH, and two shield traversing pipes (fig. 9)	4.25	0.26	93.7	45.3
SNAP 8 DR shield [16]	USAEC	same as SNAP 8 two-region DRM-1 shield, except converted to a single region shield (fig. 10)	14.93	0.89	94.2	45.6
Experimental shield no. 3 [17]	USAEC	annular shield with conical side, containing 120 lb LiH and a honeycomb matrix (fig. 11)	2.57	0.18	93.1	45.0
Segmented annular shield [18]	NASA **	four segments containing 350 lb LiH and a honeycomb matrix (fig. 12)	7.73	0.39	94.0	45.5
Thermal conductiv-ity test specimen [19]	NASA	flat disc shape, 30 in. dia. by 6 in. thick, containing 100 lb LiH and a honeycomb matrix (fig. 13)	2.45	0.13	94.6	45.8

* USAEC funded.
** NASA funded.

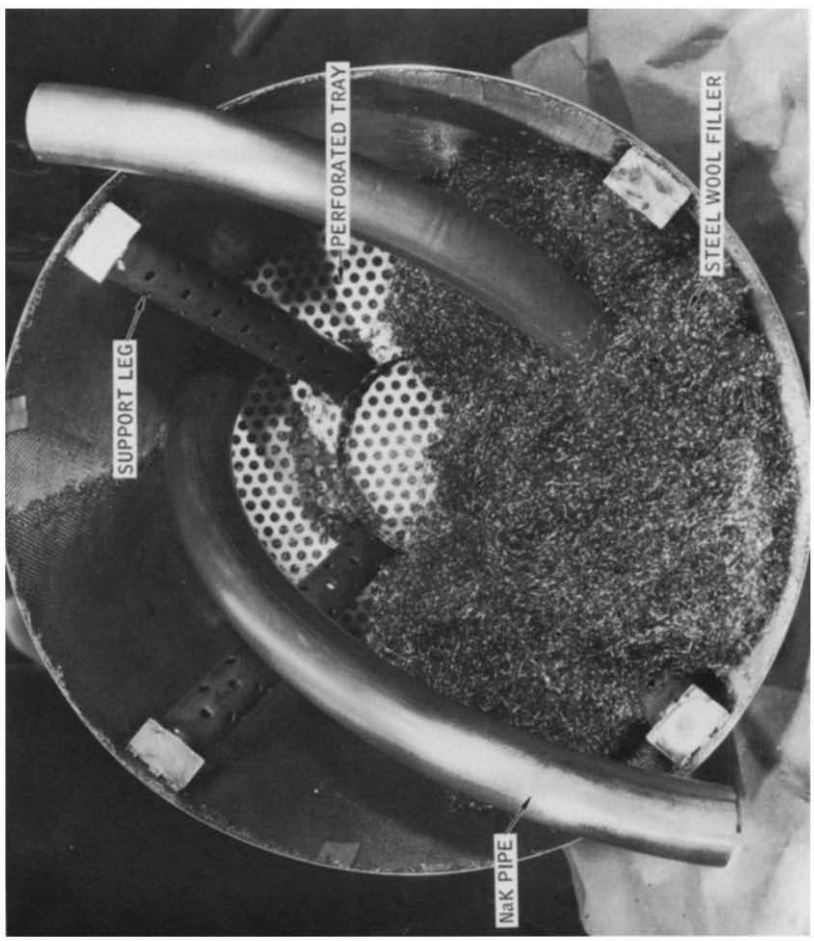


Fig. 9. Experimental shield no. 2 – view from bottom during assembly.

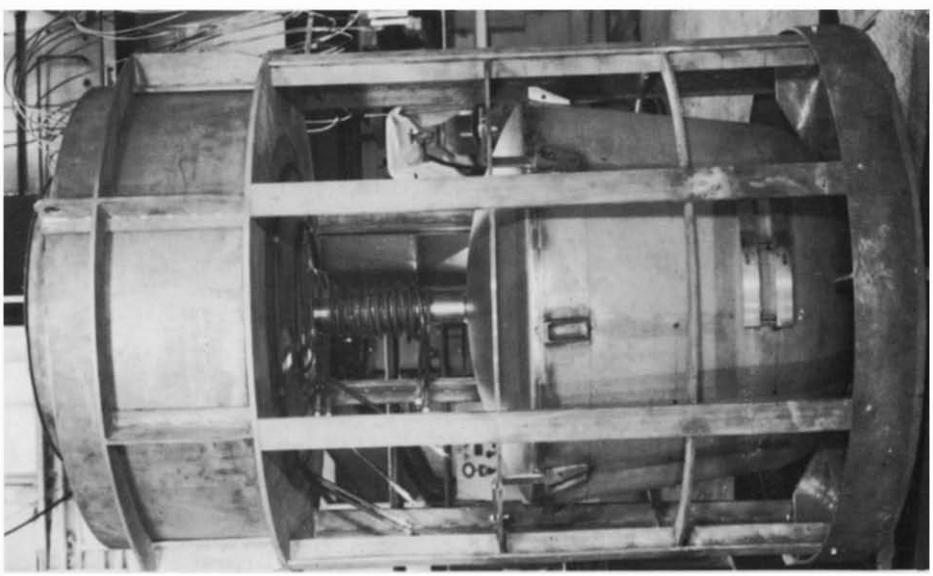


Fig. 8. SNAP 2 shield in casting support hardware.

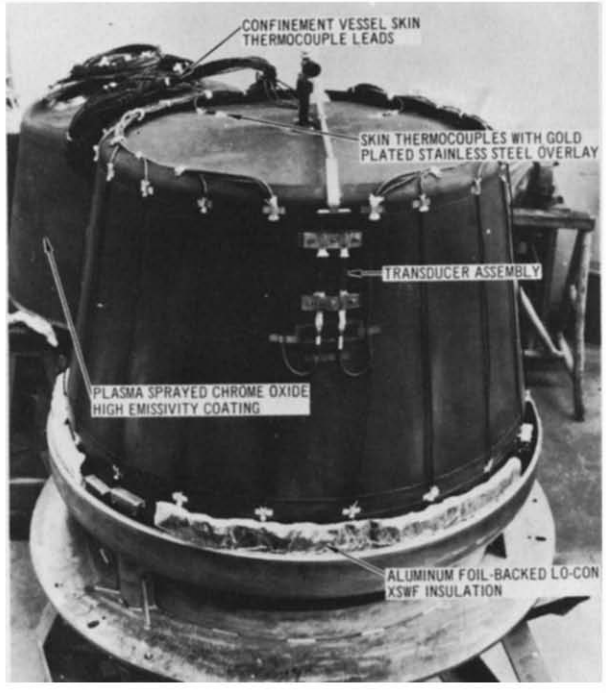


Fig. 10. Instrumented SNAP 8 DR shield.

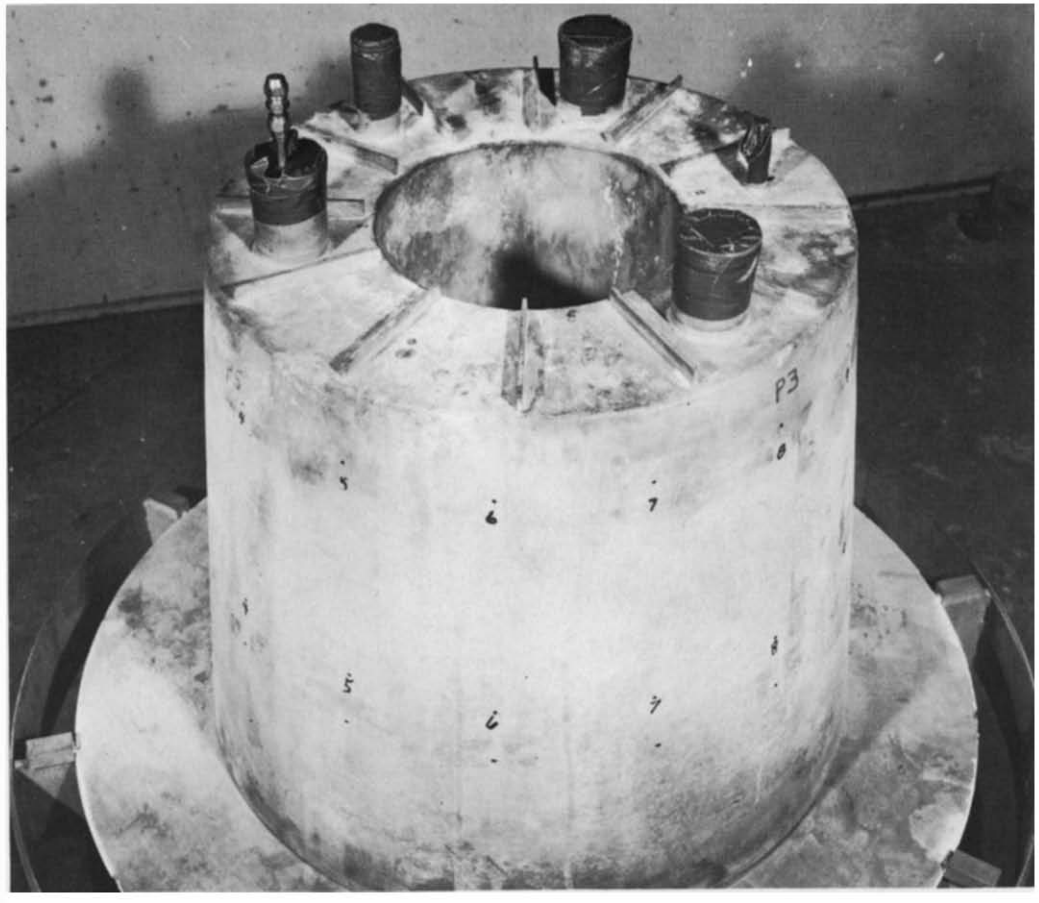


Fig. 11. Experimental shield no. 3 temporarily sealed for void volume measurement.

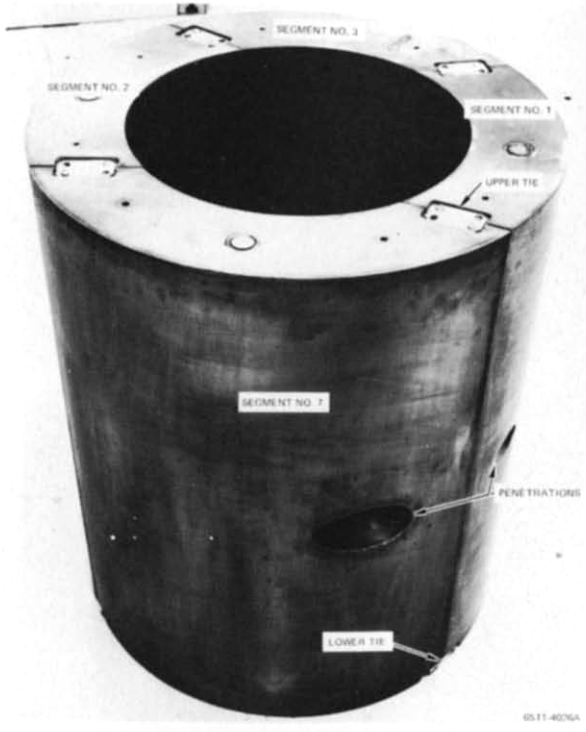


Fig. 12. Assembled segments of the NASA-Lewis Research Center zero power reactor annular shield.

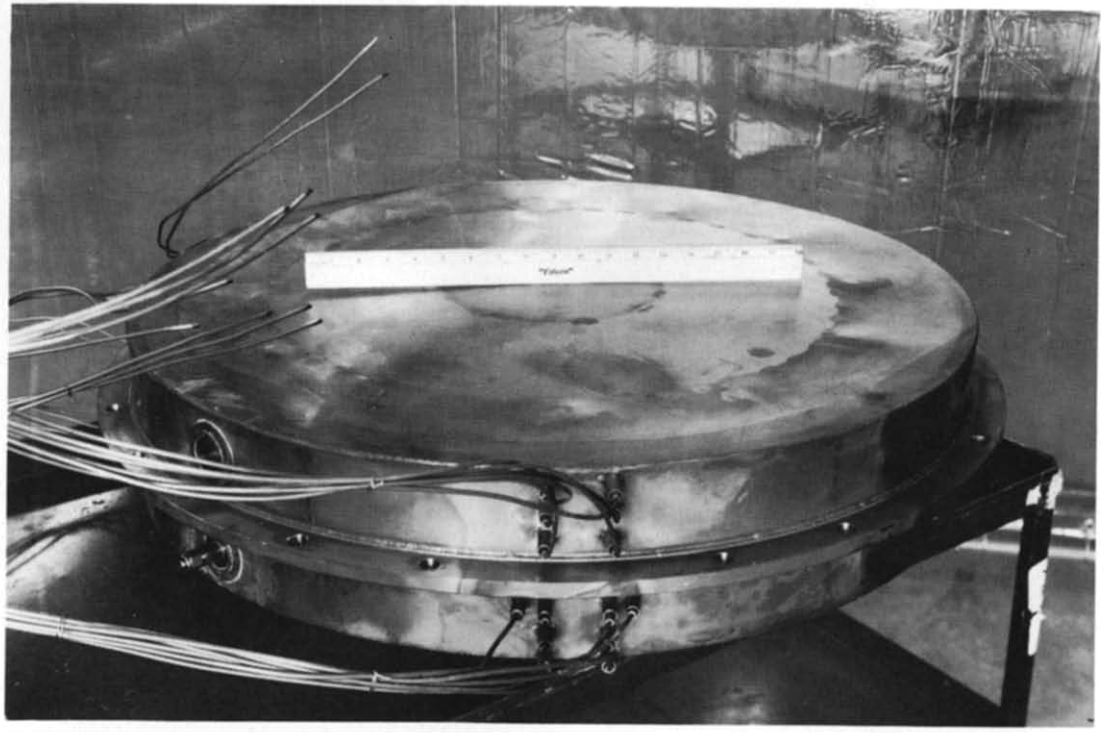


Fig. 13. NASA-Lewis Research Center LiH thermal conductivity test specimen.

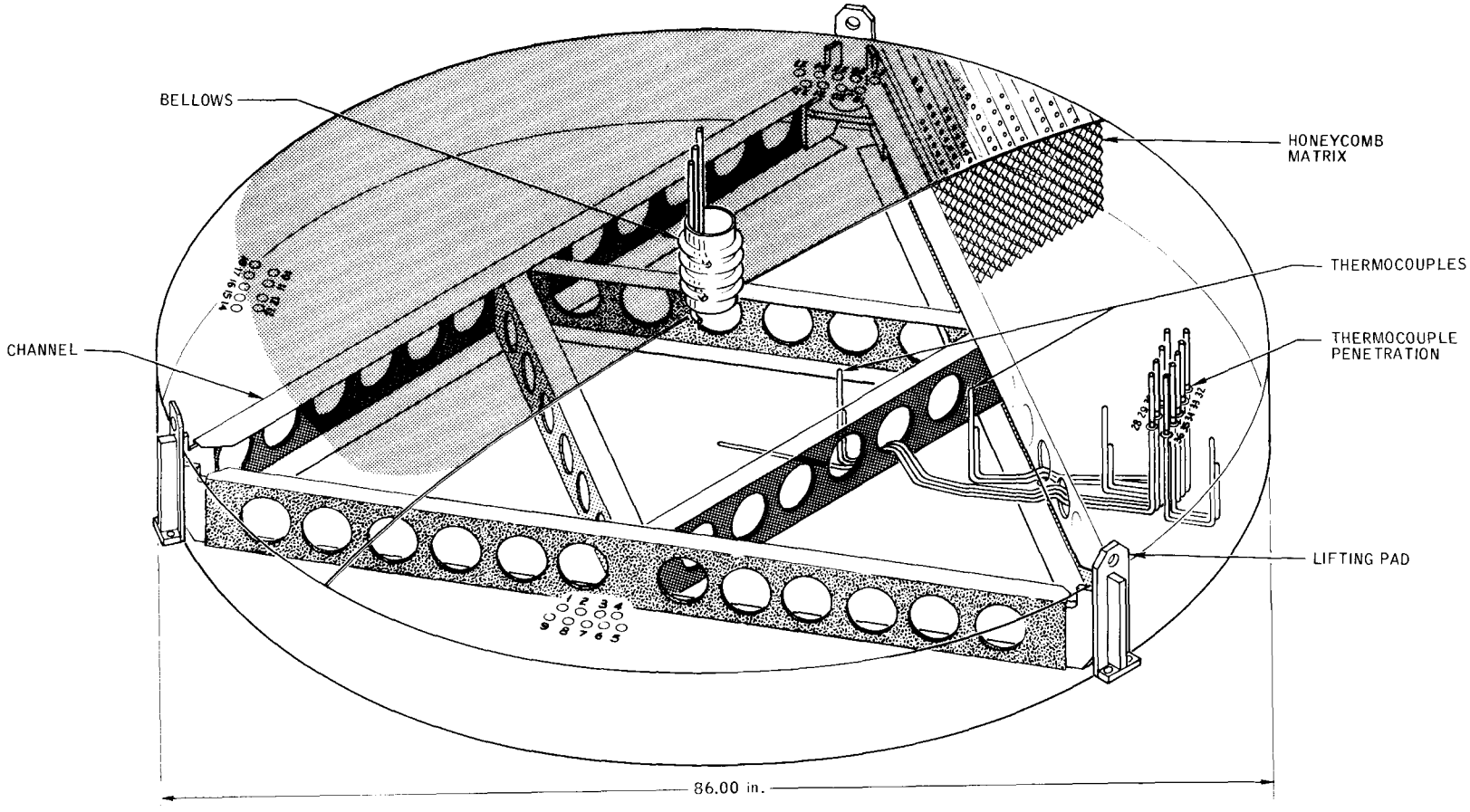


Fig. 14. Experimental shield no. 4 – the checkout shield.

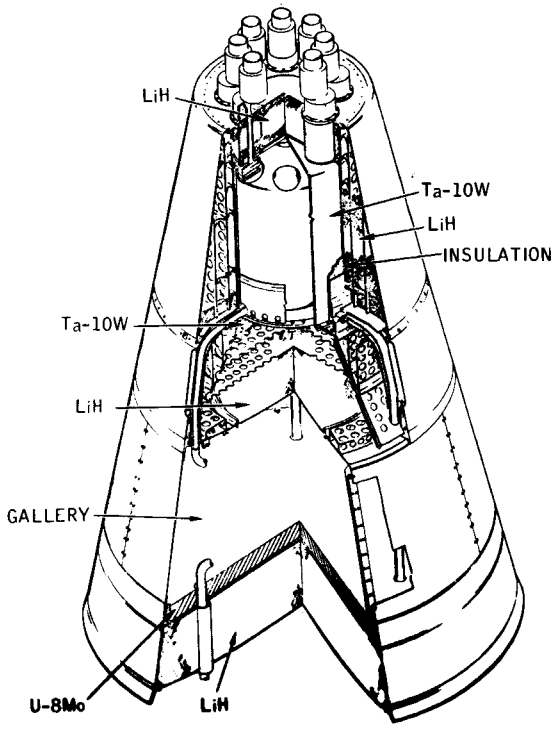


Fig. 15. Reference ZrH reactor shield system.

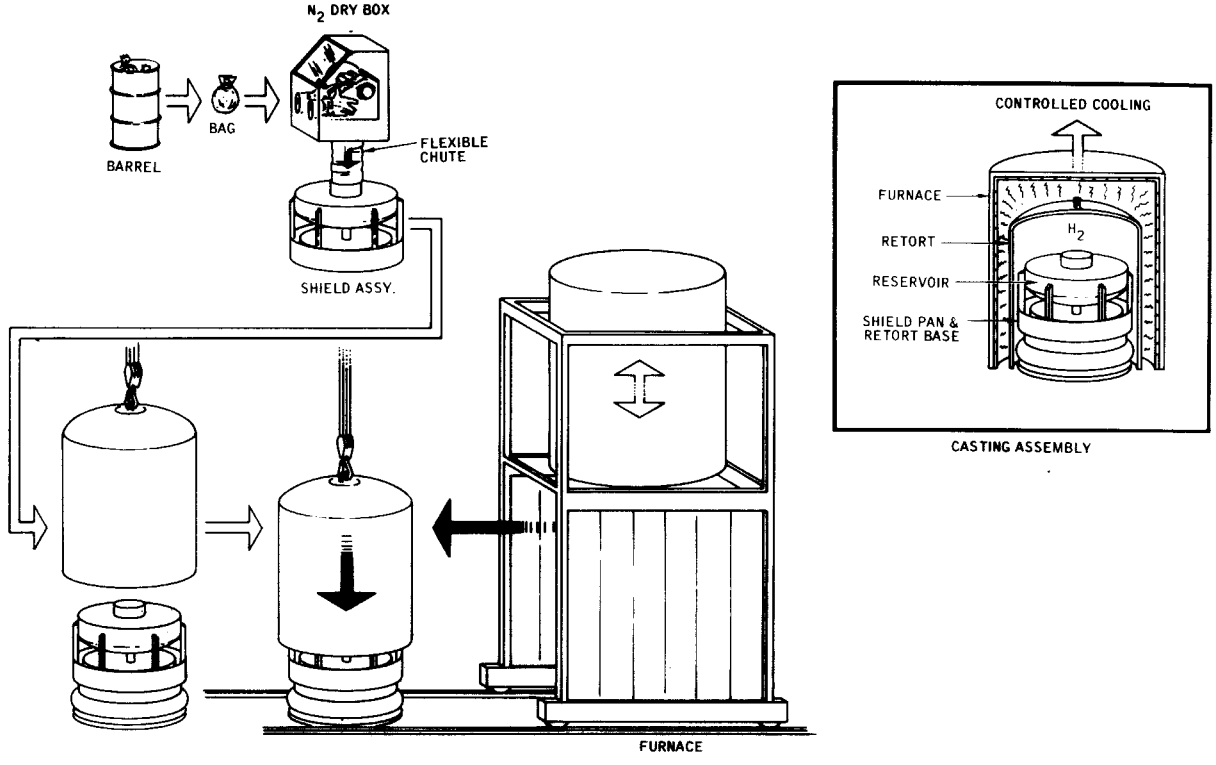


Fig. 16. LiH shield filling and casting steps.

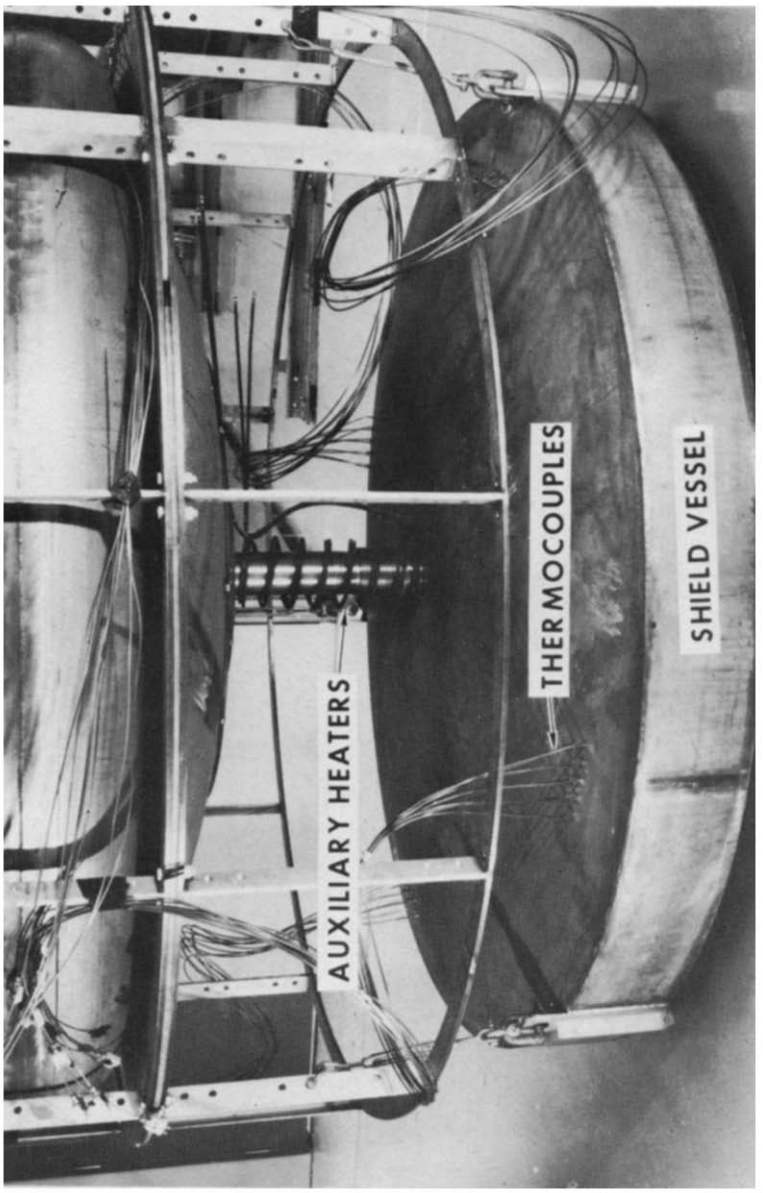


Fig. 17. Shield vessel ready to be installed in the safety container.

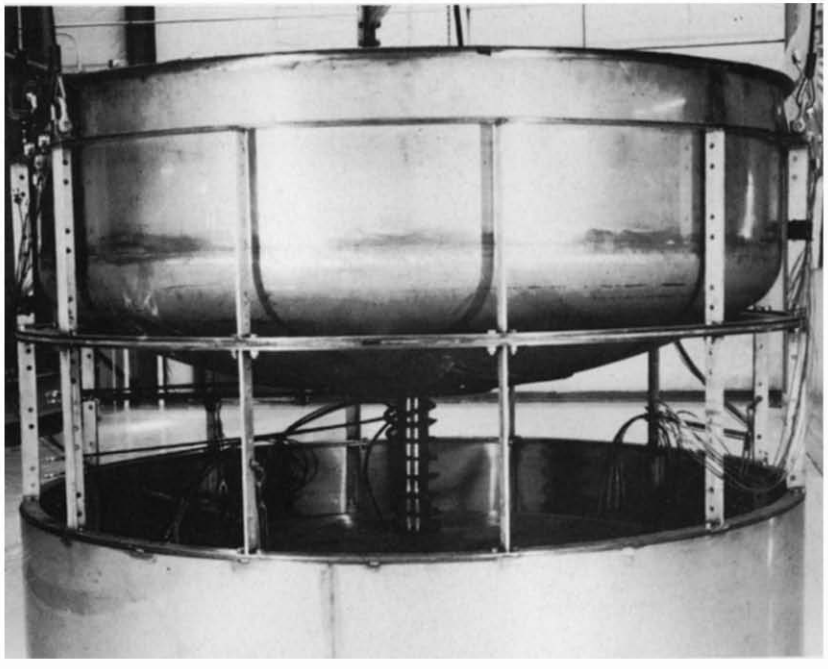


Fig. 18. Shield assembled in the casting hardware.

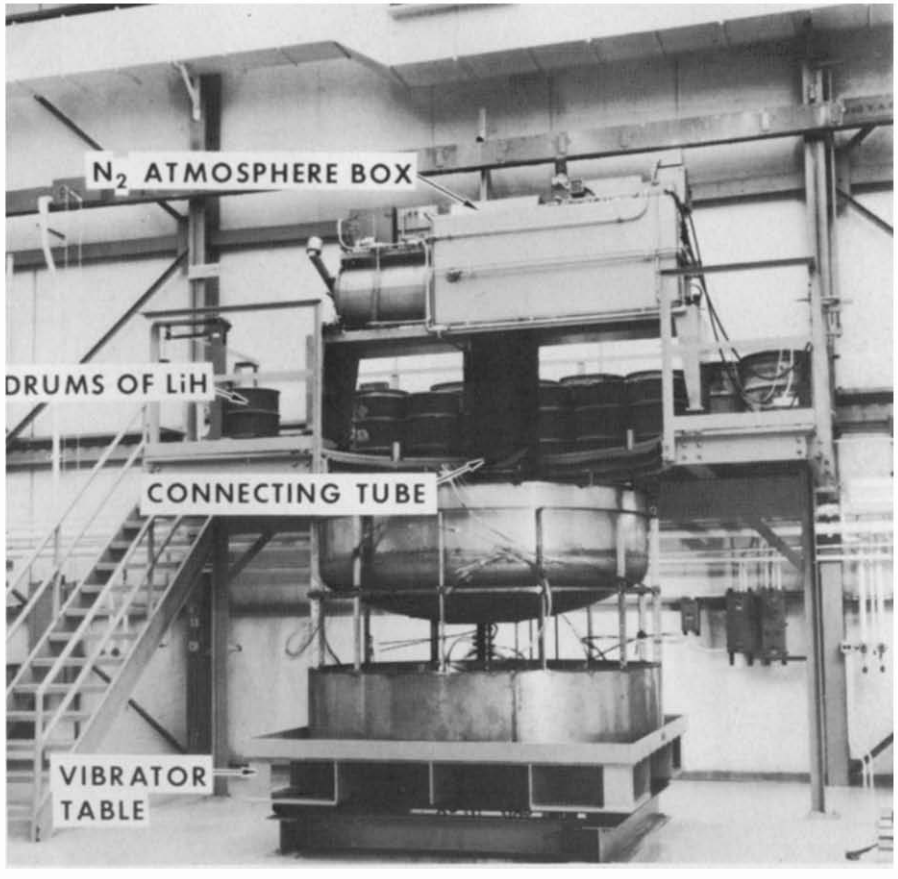


Fig. 19. Shield-casting hardware assembly on the vibrator table.



Fig. 20. Loading bags of LiH into the dry box antechamber.

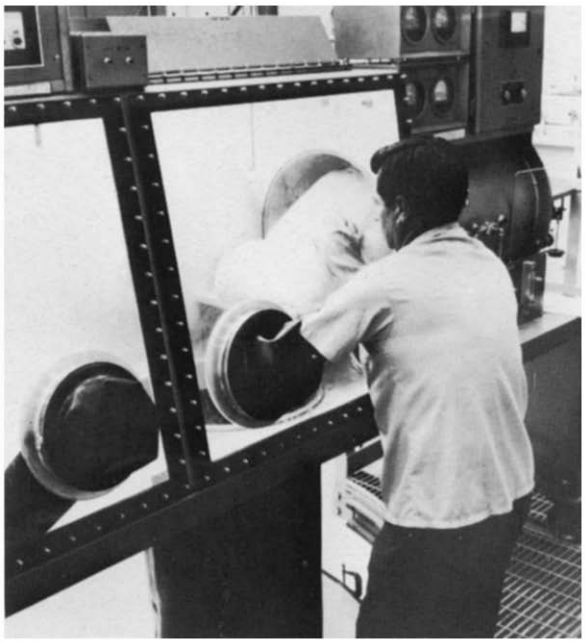


Fig. 21. Transferring bags of LiH into the dry box.

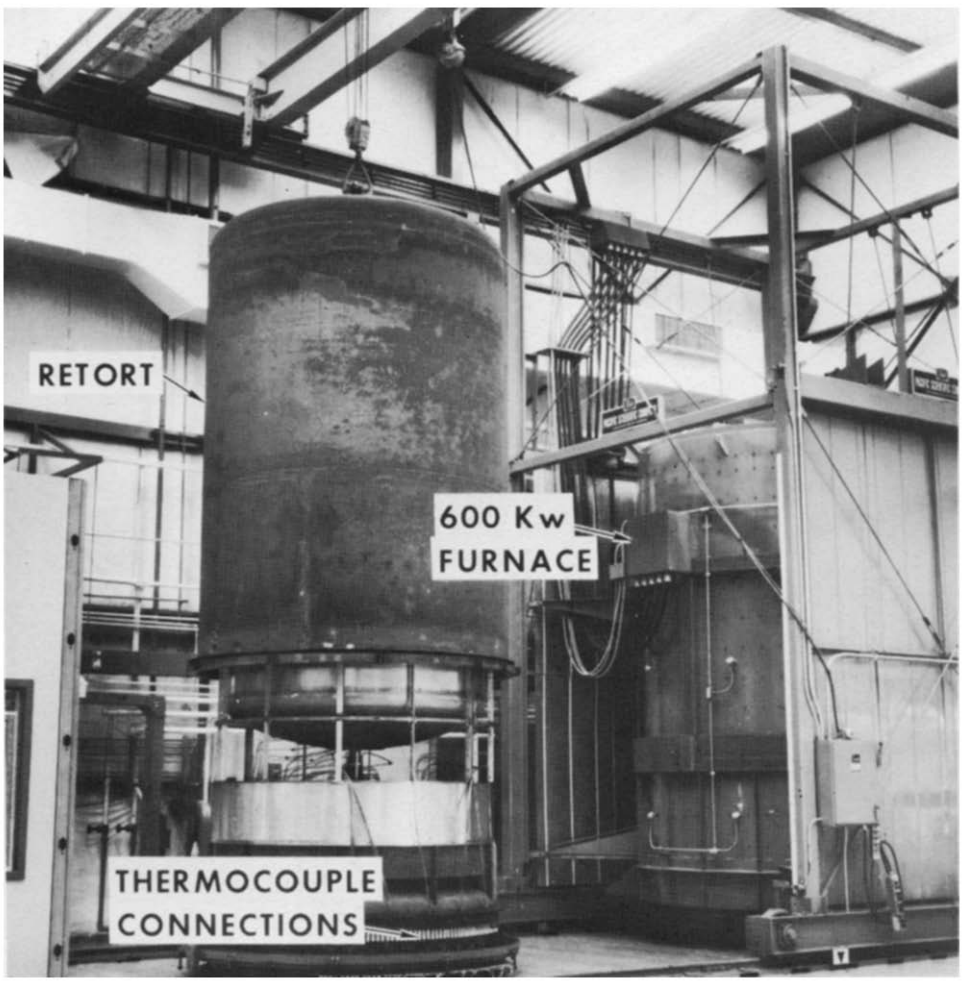


Fig. 22. Installing the retort over the shield-casting hardware assembly.

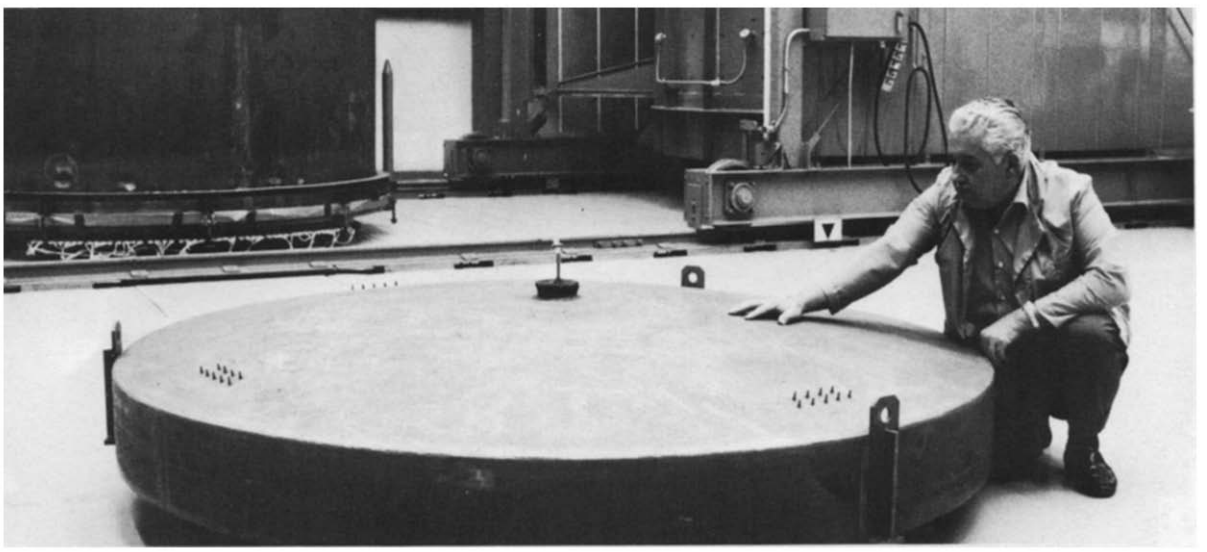


Fig. 23. The completed large LiH reaction shield.

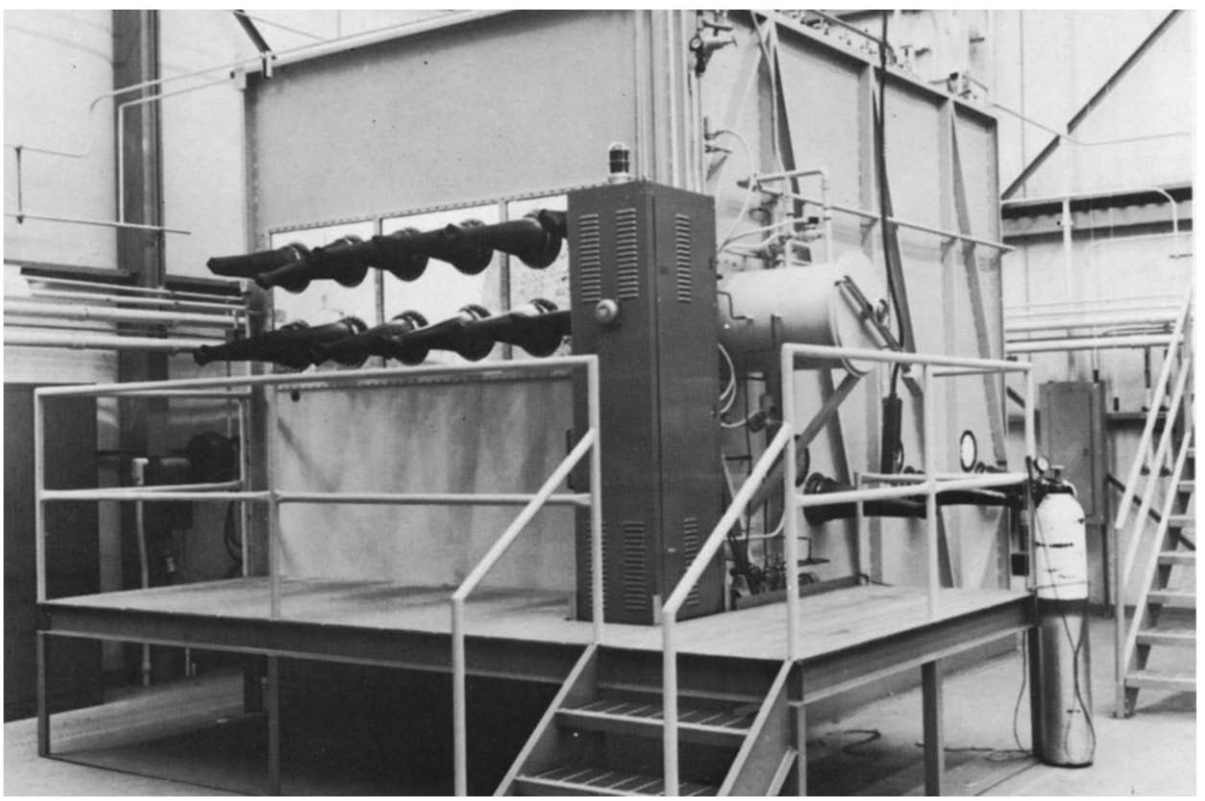


Fig. 24. The N_2 atmosphere shield examination room.



Fig. 25. The checkout shield with LiH removed from one sector.

would be faster, cheaper, and structurally more reliable than the method using a cold-pressed block of LiH fitted into a metallic container, as cited by Kurzeka [6]. Work was begun in late 1961 on a casting facility for SNAP shields at the Santa Susana Nuclear Field Laboratory, Atomics International (AI) Division of North American Aviation (now Rockwell International Corporation). In mid-1962 the facility was completed and casting operations began.

Table 1 summarizes the shields and test specimens cast in the SNAP Shield Casting Facility, as documented by Welch [15–19], and tabulates their density characteristics. The facility consists of the following special equipment:

(1) a powder-loading dry box used for loading high purity, granular LiH into the shield vessel under a controlled, protective atmosphere of nitrogen containing a maximum of 3% O_2 and 10 ppm water vapor;

(2) a retort and retort base to hold the shield during the casting cycle under an atmosphere of H_2 to prevent the dissociation of the LiH at the melting temperature; and

(3) a large, bell-type furnace (150 KW, 3 zone control) capable of heating the retort, shield, and LiH to about 1400°F.

2.1.2. Large LiH shield fabrication

The recent emphasis on the manned exploration of space, and the resulting large electrical power requirements, directed the shield fabrication efforts towards the production of larger neutron shields beyond the capability of the existing casting facility.

Recently, a new 3600 sq. ft facility was completed at the AI Nuclear Field Laboratory, designed specifically for the fabrication of future large LiH shields by the melting and casting process. Welch [20] stated that the facility is the largest in the world and is capable of producing LiH neutron shields up to 8 ft in dia. and weighing up to 10 000 lb. Table 2 compares the capability of the older, smaller facility with the new facility.

The first neutron shield fabricated in this new facility was a large, pancake shape, 86 in. dia. and 13 in. thick at the center, containing about 1700 lb of LiH. Fig. 14 is an artist's interpretation of the large shield. As shown, the shield is inverted from its normal attitude to permit the filling of the vessel to be done with the domed surface on top. Fig. 15 is a schematic view of the shield at the base of the proposed reference ZrH reactor/shield system. Fig. 16 illustrates the steps in the loading, melting, and freezing cycle to fabricate the shield, and figs. 17–22 show views of various steps of the operation.

Radiographic examination of the completed shield (fig. 23) revealed that the casting was the soundest yet produced at AI, exhibiting less than average of the normally small cracks and fissures between the LiH crystals. A series of void volume measurements were made by evacuating the shield, backfilling it to atmospheric pressure with helium from a tank of

Table 2.
Neutron shield casting capabilities.

	Old Facility	New Facility
Retort internal diameter (in.)	48	105
Retort height (in.)	60	125
Shield diameter, maximum (in.)*	42	96
Shield height, maximum (in.)*	40	80

* Maximum diameter and maximum height cannot be used in the same assembly.

known volume and pressure, and calculating the per cent void using the perfect gas law. A void volume of 5% was obtained, which is equivalent to a 95% dense shield (bulk density = 46.0 lb/ft^3).

It is reported further by Welch [21] that the large LiH shield was transferred to a specially designed and built N_2-atmosphere shield examination room (fig. 24). An extensive destructive examination, wherein the stainless steel dome was cut from the shield, and the LiH body sectioned and examined for defects, large cracks, and casting voids, confirmed the soundness and uniformity of the large casting (fig. 25).

3. Summary

The LiH casting development has progressed during the last one and a half decades from small, laboratory-type castings to massive castings of near-ton size. This casting development work also has experienced a logical sequence of increasing complexity resulting in a technology capable of supplying large LiH neutron shields for current and forseeable future needs.

References

[1] F.H. Welch et al., Lithium hydride technology I. Properties of lithium hydride and corrosion studies, NAA-SR-9400, 8 May (1964).
[2] F.H. Welch, Lithium hydride technology III. Properties of lithium hydride for SNAP shielding applications, NAA-SR-9400, III, 1 May (1967).
[3] C.W. Hamill et al., Metal hydrides for shielding applications, Y-1366, 6 October (1961).
[4] F.L. Woods et al., Lithium hydride as a thermal energy storage material, ASD-TR-61-427, 29 August (1961).
[5] W.G. Baxter and F.H. Welch, Shield materials, APEX-915, 1 March (1962).
[6] W.J. Kurzeka, SNAP 10A component development summary, volume 3 – shield, ground test assembly and materials application, NAA-SR-9898, 3, 10 February (1965).
[7] D.H. Schell et al., Fabrication of lithium hydride by hot pressing, LAMS-2108, June (1958).
[8] ANL Staff, Report for July, August and September, Experimental Nuclear Physics Division and Summary Report for April through September, ANL-4208,4 October (1948).
[9] J.M. West, Induced activity in lithium hydride, HW-10090, January (1948).
[10] E.A. Aitken and M. Tetenbaum, Lithium hydride report, DC-56-7-136 (Declassified) 19 July (1956).
[11] F.H. Welch, Summary report on the lithium hydride development program for project 103, TID-6543 (DC-59-8-181), 6 August (1959).
[12] W.G. Baxter and F.H. Welch, High temperature, light weight neutron shielding for nuclear space power plants, TID 7653, Pt II, 15 August (1962).
[13] F.H. Welch and J.G. Attanas, Lithium hydride casting development program – side shield, full scale D140E1 – milestone reports I and II on the casting of pipe and prototype specimens, DC 61-1-85, 23 January (1961).
[14] R.H. Poor, Full scale D140E1 shield casting program, APEX-740, August (1961).
[15] F.H. Welch, Lithium hydride technology II: Fabrication of shadow shields for SNAP applications, NAA-SR-9400, II, 30 September (1965).
[16] F.H. Welch, Lithium hydride technology V: Testing and examination of SNAP shadow shields, NAA-SR-9400 V, 31 August (1967).
[17] F.H. Welch, Lithium hydride technology: Design, casting, and examination of an annular LiH shield (experimental shield #3) AI-AEC-TDR 12905, 3 December (1969).
[18] F.H. Welch, Lithium hydride shield segment fabrication – final report, NASA-CR-120782, 3 September (1971).
[19] F.H. Welch, Lithium hydride thermal conductivity test specimen final report, TI-001-230-004, 22 May (1972).
[20] F.H. Welch, Space reactor shielding fabrication, Proceedings of the National Symposium on Natural and Manmade Radiation in Space, 1–5 March, 1971, E.A. Warman (Ed.), NASA-TM-X-2440, January (1972).
[21] F.H. Welch, Lithium hydride technology: design, fabrication, casting, and destructive examination of experimental shield #4, the LiH checkout shield, TR-696-230-006, 24 June (1971).